

Database Design And Summary

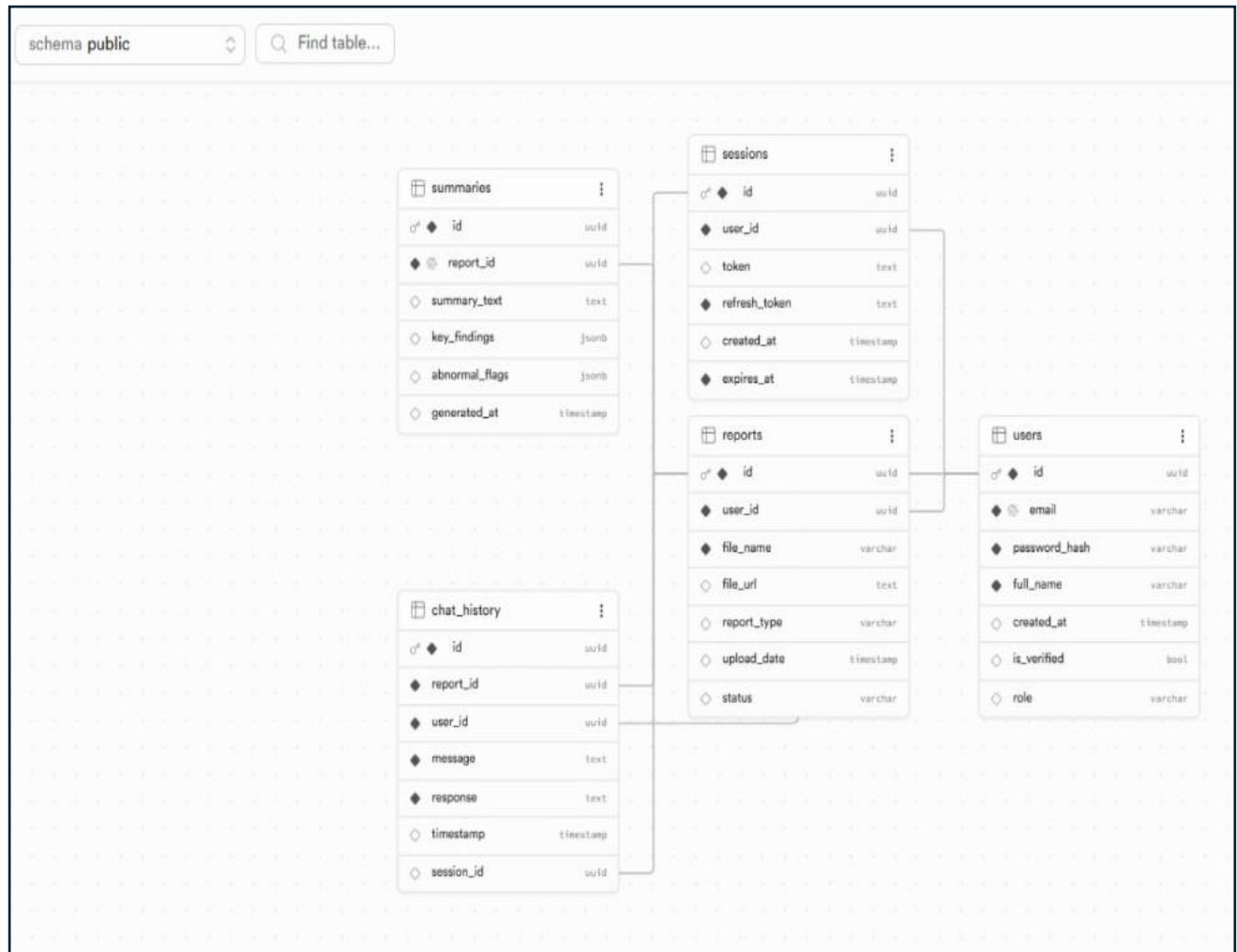


Fig: Database Design

Database Design Summary

Here's a concise breakdown of your database design:

Core idea

This schema supports a user-based reporting + AI/chat system, where users upload reports, generate summaries, and interact via chat sessions tied to those reports.

1. Users

Stores account information.

- id (UUID, PK)
- email, password_hash
- full_name
- created_at
- is_verified
- role

Purpose: central identity table for authentication and access control.

2. Sessions

Manages login/auth sessions.

- id
- user_id (FK → users)
- token, refresh_token
- created_at, expires_at

Purpose: handles authentication and session lifecycle.

3. Reports

Represents uploaded files/documents.

- id
- user_id (FK → users)
- file_name, file_url
- report_type
- upload_date
- status

Purpose: core content entity that everything else links to.

4. Summaries

AI-generated or processed insights from reports.

- id
- report_id (FK → reports)
- summary_text

- key_findings (JSONB)
- abnormal_flags (JSONB)
- generated_at

Purpose: stores structured analysis results per report.

5. Chat History

Stores conversation logs tied to reports and sessions.

- id
- report_id (FK → reports)
- user_id (FK → users)
- session_id (FK → sessions)
- message
- response
- timestamp

Purpose: tracks user interactions with the system (likely AI Q&A per report).

Relationships (big picture)

- Users → Reports (1-to-many)
- Users → Sessions (1-to-many)
- Reports → Summaries (1-to-many or 1-to-1 depending on usage)
- Reports → Chat History (1-to-many)
- Sessions → Chat History (1-to-many)